

BIH-VI Modules A-B Addendum

Bispectrum Stress Test and Transfer-Scale Mapping for the Sqrt-Reg EFT-Clock Branch

Status: technical addendum to the BIH-VI report

Abstract

This addendum extends the BIH-VI technical report by documenting two post-freeze modules. Module A evaluates the non-Gaussianity risk of the BIH-VI exit switch using the dominant switch-enhanced cubic interaction. A high-resolution equilateral in-in estimate confirms that CMB-pivot-side modes remain effectively Gaussian, while exit-scale modes develop a localized, large normalized feature amplitude coincident with a deep scalar-power suppression. Module B maps the exit feature to present-day physical scales and shows that, for $N_* = 55$, the switch feature lies at ultra-small comoving scales, $k_{\text{peak}} \sim 10^{23} \text{ Mpc}^{-1}$, corresponding to a present-day frequency of order 10^8 Hz . The feature is therefore not a CMB or LSS target; the primary CMB-facing BIH-VI predictions remain $n_s \sim 0.965$, $r \sim 10^{-3}$, and $\alpha_s \sim -6e-4$. The exit feature is retained as a secondary ultra-small-scale relic prediction.

1. Executive Summary

- BIH-VI remains a stable EFT-clock realization of the Sqrt-Reg entropy background, not a final microscopic substrate theory.
- The benchmark $p=3$, $\mu=1$, $c_s=1$, $N_*=55$, $z_e=1/3$ gives $n_s=0.9652$, $r=1.4e-3$, $\alpha_s=-5.8e-4$, and $H_c=3.8e-6 \text{ M}_{\text{Pl}}$.
- The exit switch with $\Delta=0.01$ is pivot-safe and numerically stable in the Mukhanov-Sasaki stress test.
- The leading equilateral feature-bispectrum estimate gives $|f_{\text{NL}}| \sim 3e3$ at $K \sim 3.58$, but only where the two-point power is suppressed by $\sim 3e-5$.
- The absolute bispectrum proxy is smooth and finite; the large normalized f_{NL} mainly reflects the small denominator in the conventional f_{NL} normalization.
- The exit feature maps to $k_{\text{peak}} \sim 1.3e23 \text{ Mpc}^{-1}$, or $f_{\text{peak}} \sim 2e8 \text{ Hz}$; it is not a CMB-scale feature.

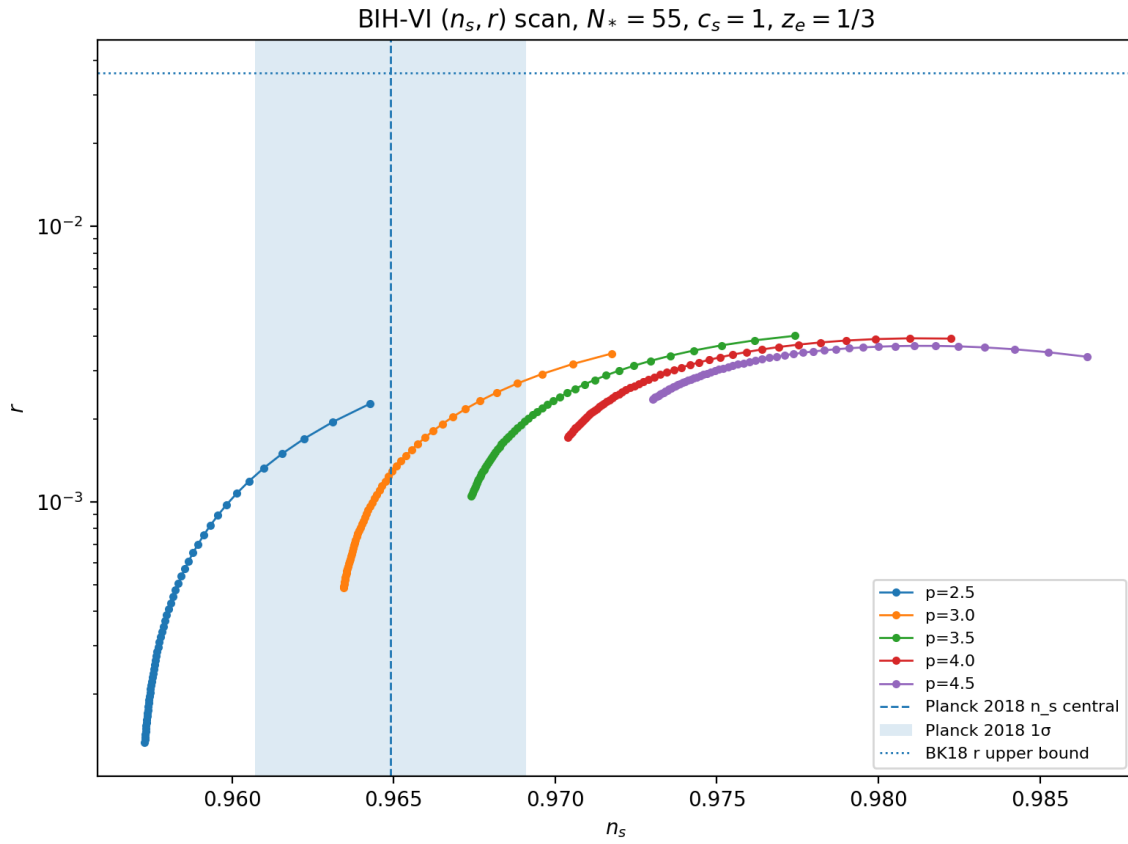


Figure 1. BIH-VI parameter scan in the (n_s, r) plane. The viable island is centered near $p=3, \mu \sim 1$.

2. Core BIH-VI Benchmark Equations

The stable EFT-clock branch is specified by the Sqrt-Reg background and the gear profile:

$$H^2 = H_c^2 y(2-y), \quad z = 1-y, \quad m(y) = \mu z^p.$$

$$\epsilon_H = m(y)(1-y)/(2-y).$$

$$S_{\text{zeta}}^{(2)} = M_{\text{Pl}}^2 \int dt d^3x a^3 [\epsilon_H/c_s^2] [\dot{z}^2 - c_s^2 (\text{grad } z)^2/a^2].$$

Benchmark: $p=3, \mu=1, c_s=1, N_*=55, z_e=1/3$.

Predictions: $n_s=0.9652, r=1.4e-3, \alpha_s=-5.8e-4, H_c=3.8e-6 M_{\text{Pl}}$.

3. Module A: Exit-Switch Bispectrum and Non-Gaussianity

Module A tests whether the exit switch produces dangerous non-Gaussianity. The calculation isolates the dominant switch-enhanced cubic interaction, which is controlled by the rapid variation of $\eta_H = d \ln \epsilon_H / dN$ during the transition.

$$S_3^{\text{switch}} \supset (M_{\text{Pl}}^2/2) \int dt d^3x a^3 \epsilon_H H \eta_N z^2 \dot{z}, \quad \eta_N = d \eta_H / dN.$$

$$H_{\text{int}}^{\text{switch}} \approx -(M_{\text{Pl}}^2/2) \int dt d^3x a^3 \epsilon_H H \eta_N z^2 \dot{z}.$$

This is a leading-interaction feature estimate. A full all-operator EFT bispectrum and all triangle configurations remain outside the scope of this addendum.

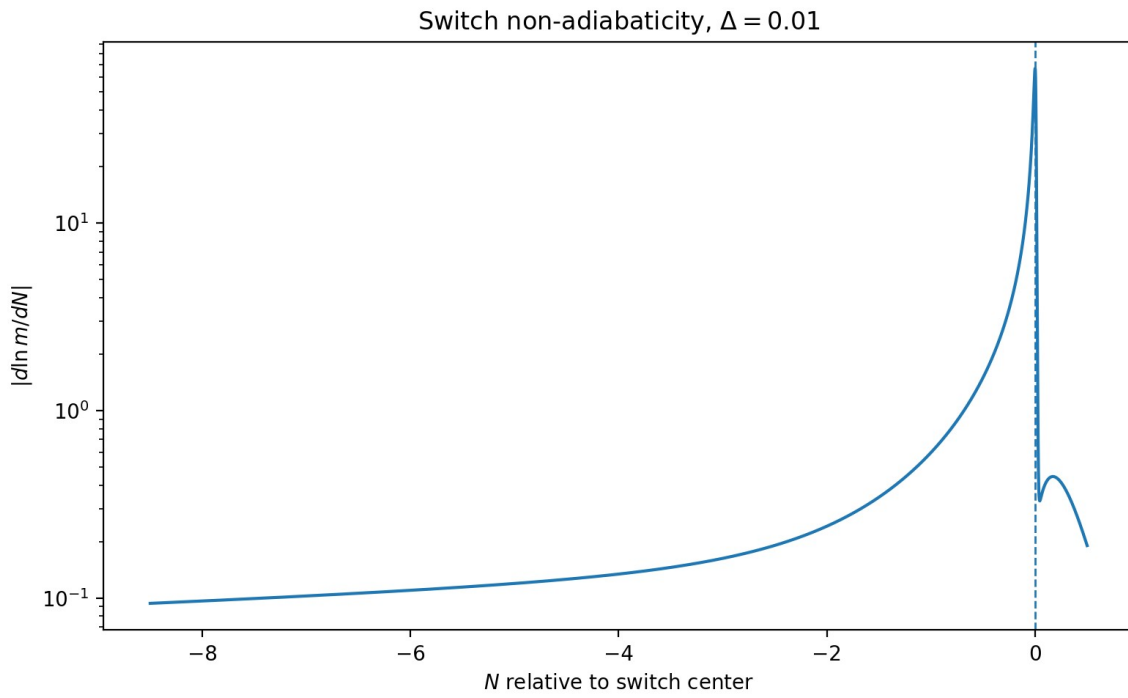


Figure 2. Switch non-adiabaticity for $\Delta=0.01$. The transition is localized and sharp.

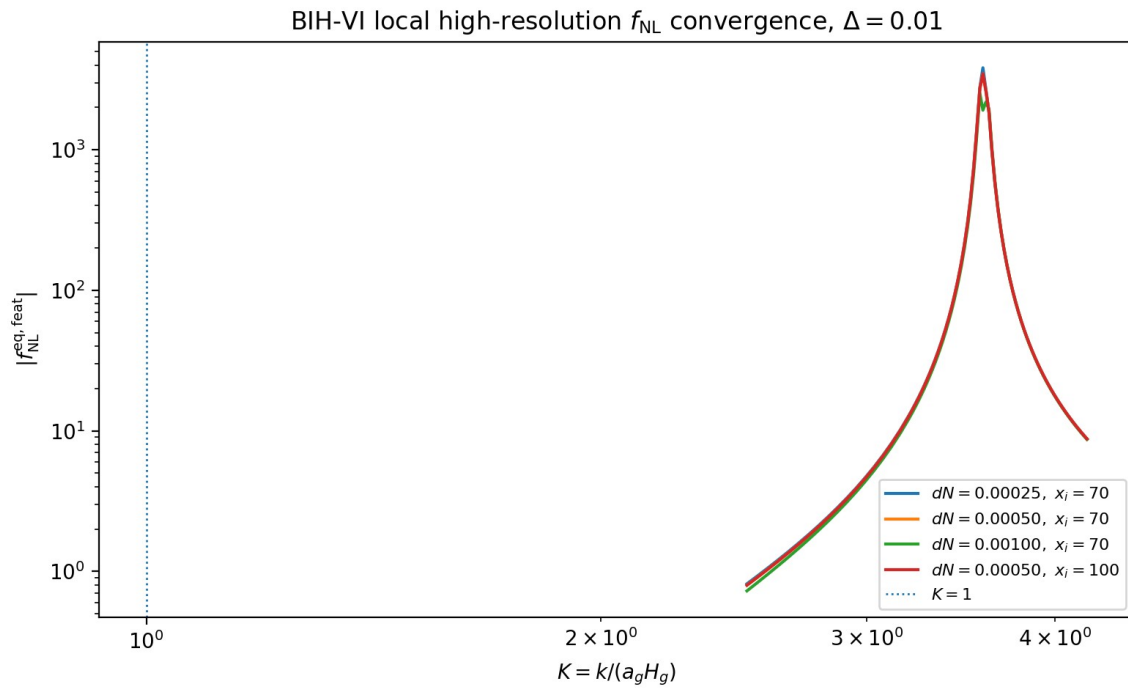


Figure 3. High-resolution local convergence of the equilateral feature f_{NL} estimate near the exit-scale peak.

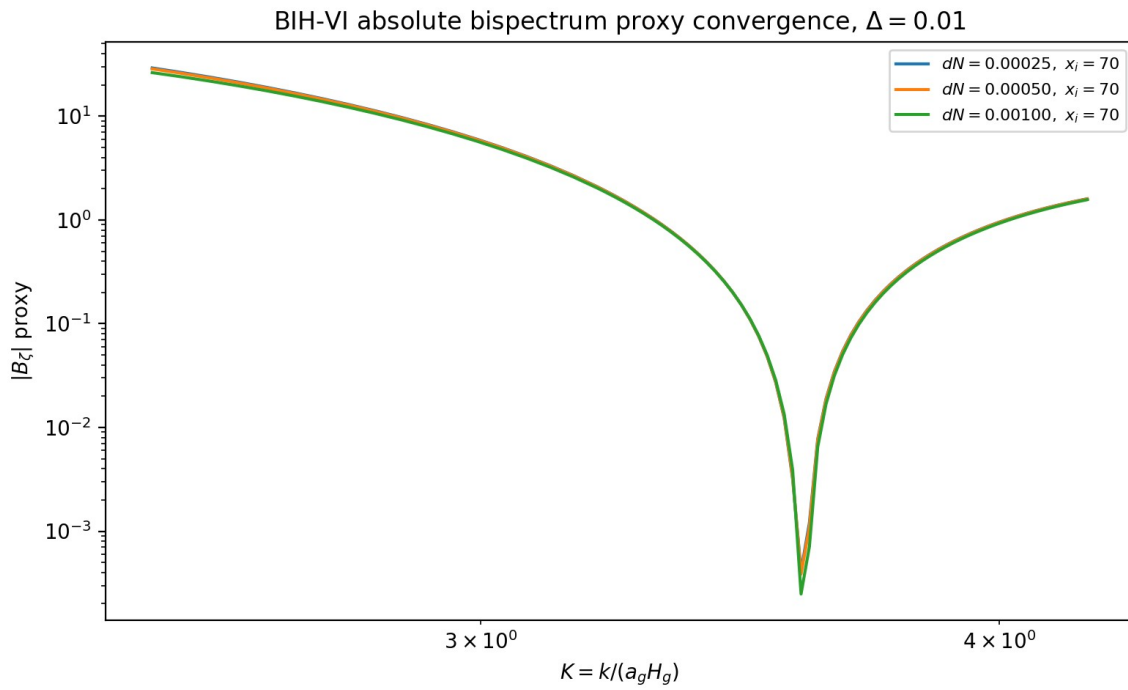


Figure 4. Absolute bispectrum proxy. The proxy remains smooth and finite even where normalized f_{NL} spikes.

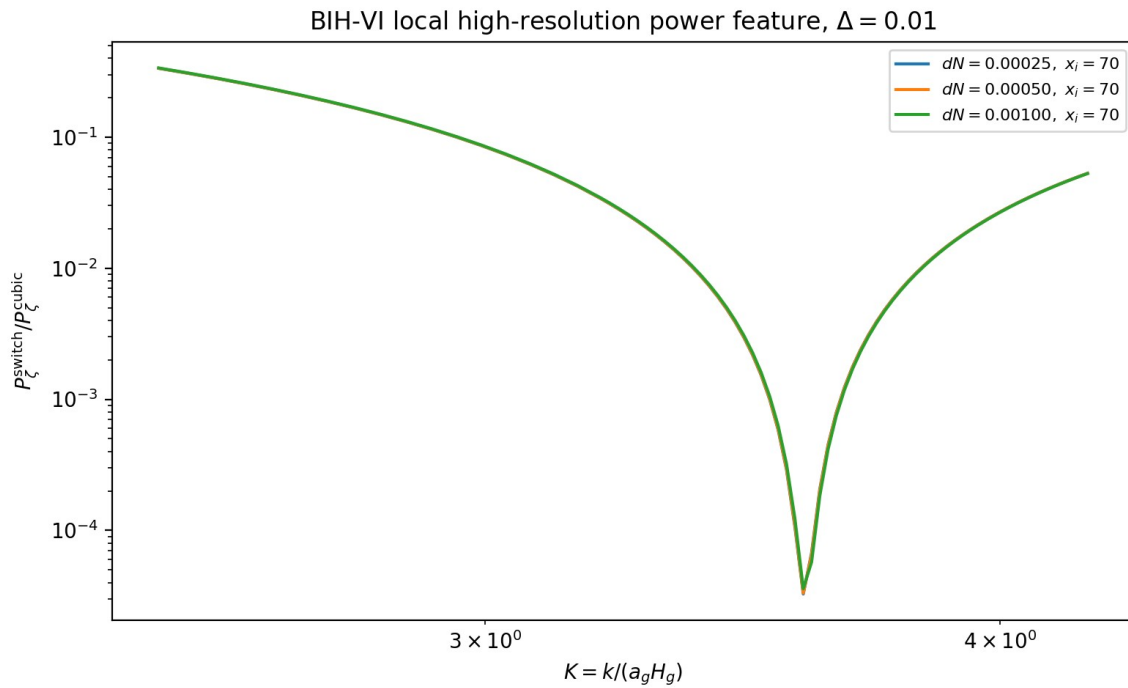


Figure 5. Local high-resolution power feature. The normalized f_{NL} peak coincides with a deep two-point suppression.

3.1 High-Resolution Feature Result

Quantity	Benchmark Value	Interpretation	Status
K_{peak}	~ 3.58	Exit-scale feature location	Converged
$ f_{\text{NL}}^{\text{eq,feat}} _{\text{peak}}$	$\sim 3.1e3$	Large normalized feature	Feature estimate

		amplitude	
P_switch/P_cubic at peak	$\sim 3.3\text{e-}5$	Deep scalar-power suppression	Converged
Absolute B proxy	smooth, finite	No absolute blow-up	Pass
$K < 1$ f_NL	$O(0.1)$ or smaller	Pivot side effectively Gaussian	Pass

3.2 Module A Interpretation

The exit switch produces a localized equilateral-type feature. The large normalized feature amplitude occurs only where the power spectrum is deeply suppressed. The absolute bispectrum proxy does not diverge, so the result should be described as a localized exit-scale feature rather than broad-band catastrophic non-Gaussianity.

Report wording: localized exit-scale power suppression + large normalized feature bispectrum; CMB-pivot modes remain effectively Gaussian.

4. Module B: Transfer-Scale Mapping

Module B maps the exit-scale feature to physical comoving wavenumber and present-day frequency. This is the decisive check against over-interpreting the splash as a CMB feature.

$$K = k/(a_g H_g), \quad k_{\text{pivot}} = 0.05 \text{ Mpc}^{-1}, \quad N_* = 55.$$

$$k_{\text{switch}} \approx 3.6\text{e}22 \text{ Mpc}^{-1}, \quad k_{\text{peak}} \approx 1.3\text{e}23 \text{ Mpc}^{-1}.$$

$$f_{\text{today}} \approx 1.54\text{e-}15 (k/\text{Mpc}^{-1}) \text{ Hz}, \quad f_{\text{peak}} \approx 2\text{e}8 \text{ Hz}.$$

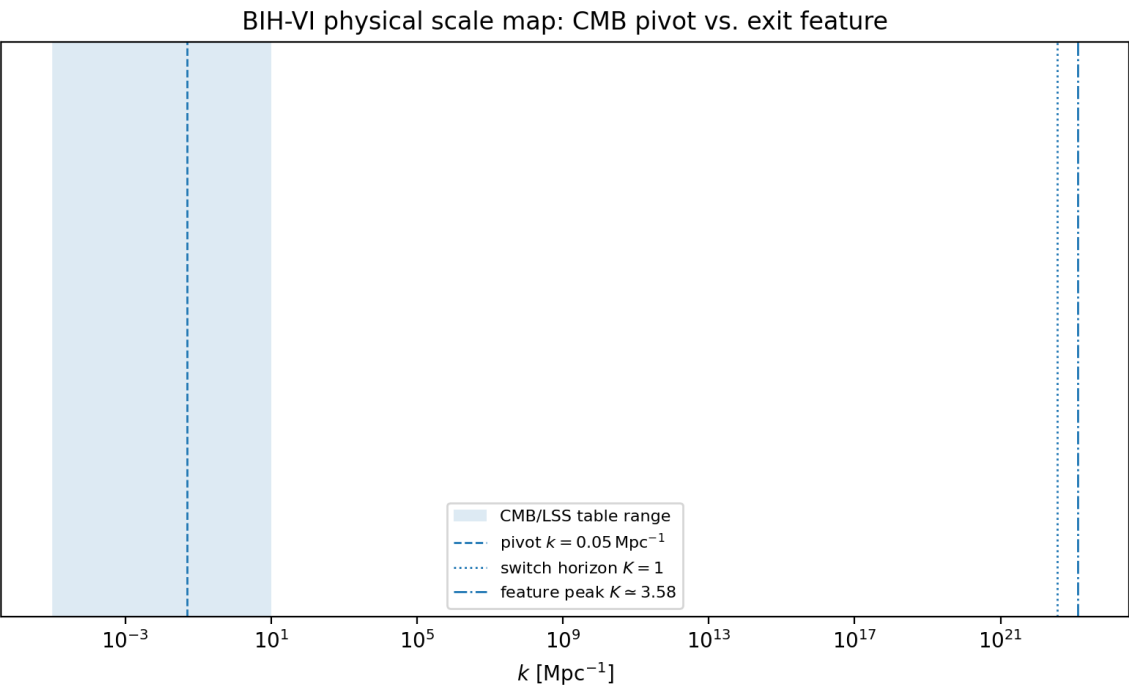


Figure 6. Physical scale map. The exit feature is exponentially far beyond CMB and LSS scales.

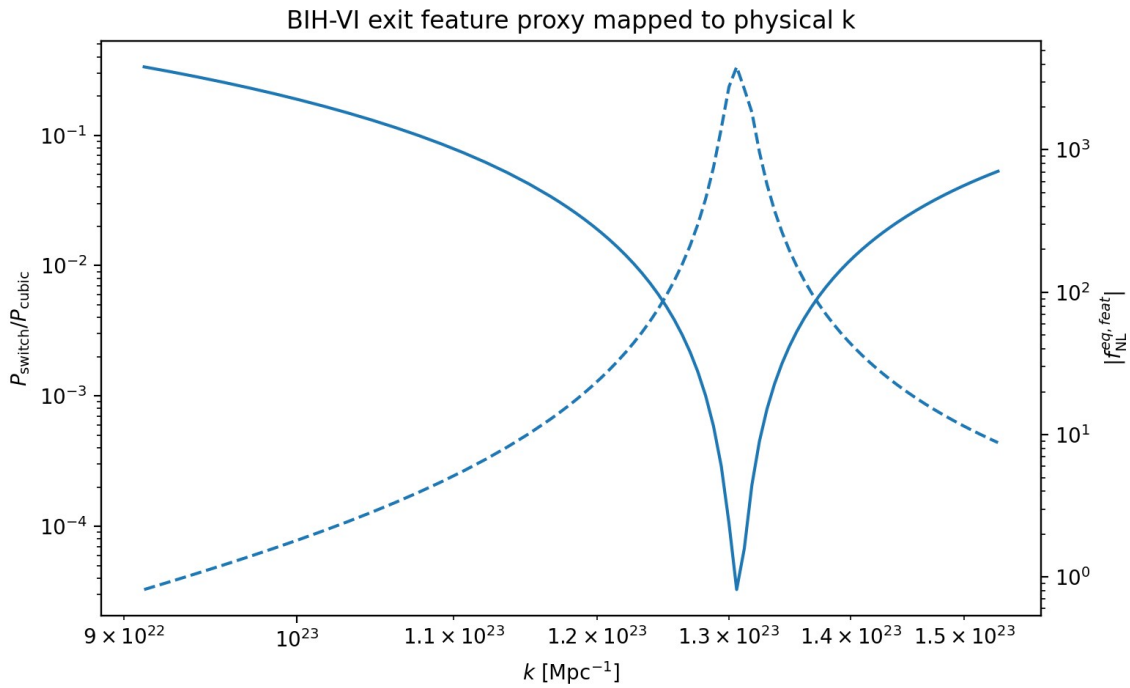


Figure 7. Exit-feature proxy mapped to physical k . The feature lies around $k \sim 1e23 \text{ Mpc}^{-1}$.

4.1 Module B Verdict

- The CMB/LSS range sees the smooth BIH-VI spectrum, not the exit splash.
- The exit feature corresponds to ultra-small comoving scales and present-day frequencies around hundreds of MHz.
- The scalar feature is power-suppressed, so it is unlikely to generate primordial black holes in the benchmark branch.
- The primary observational hooks remain n_s , r , and α_s ; the exit feature is a secondary relic.

5. Technical Translation of the Black-Oil Intuition

The black-oil image is useful as an engineering metaphor only when translated into EFT terms. The oil is not a literal substance in this report. It represents the single-clock medium defined by the EFT action.

Intuition	Technical Object
black oil / material flow	EFT-clock background trajectory $y(N)$
surface tension	Sqrt-Reg cap H_c
viscosity	sound speed and EFT operators, including M_2^4
laminar flow	stable, Gaussian pivot-scale perturbations
splash / cavitation	localized exit-scale feature in P_ζ and B_ζ

6. Scope Boundaries and Residual Uncertainties

EFT vs. microscopic completion

BIH-VI proves the existence of a stable EFT branch realizing the Sqrt-Reg background. It does not derive the final UV-complete substrate theory.

Switch-profile sensitivity

Delta=0.01 is pivot-safe and numerically stable, but the transition profile is phenomenological and not yet derived from a microscopic entropy-substrate mechanism.

Single-clock limitation

The present construction is a single-clock EFT realization. Multi-field or isocurvature effects are outside this module.

Bispectrum truncation

Module A uses the dominant switch-enhanced cubic interaction. Full all-operator bispectrum templates remain future work.

Boltzmann transfer

Module B shows the exit feature is outside CMB/LSS scales. A full Boltzmann calculation is unnecessary for the feature at $N_*=55$ but may still be useful for the smooth pivot spectrum.

7. Updated Prediction Hierarchy

Tier	Prediction	Status
Primary	$n_s \approx 0.965$, $r \approx 1.4e-3$, $\alpha_s \approx -5.8e-4$	CMB-facing
Secondary	Localized exit-scale power suppression and feature bispectrum	Ultra-small-scale relic
Tertiary	Possible high-frequency induced-GW/transition relics	Speculative; amplitude likely small

8. Conclusion

Modules A and B sharpen the BIH-VI framework without changing the core branch. The CMB-facing predictions remain smooth and pivot-safe. The exit switch produces a genuine localized feature, but it lies at ultra-small scales for the benchmark $N_*=55$. The correct final framing is therefore: BIH-VI contains a stable Sqrt-Reg EFT-clock branch with a viable CMB parameter island and a controlled ultra-small-scale exit relic.

Appendix: Artifact Index

- bih_vi_fnl_local_highres_convergence.png
- bih_vi_absB_local_highres_convergence.png
- bih_vi_power_local_highres_convergence.png
- bih_vi_fnl_local_highres_scan.csv
- bih_vi_fnl_local_highres_summary.csv
- bih_vi_fnl_local_highres_diffs.csv
- bih_vi_module_b_scale_map.png
- bih_vi_module_b_exit_feature_physical_k.png

- bih_vi_CLASS_primordial_cmb_range.csv
- bih_vi_primordial_exit_feature_scale_map.csv
- bih_vi_module_b_scale_mapping_summary.csv